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Subject:AIML

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Branch:electronics and computer

Q1.(Variables & Naming Rules)

```
full_name = "Vaishnavi Gayake"
age = 17
height = 1.60
is_student = True
print("Full Name:", full_name)
print("Age:", age)
print("Height:", height)
print("Student:", is_student)
print("\nData Types:")
print("full_name:", type(full_name))
print("age:", type(age))
print("height:", type(height))
print("is_student:", type(is_student))
```

Q2.(Data Types)

```
name = "Vaishnavi"    # str
age = 17              # int
height = 1.60         # float
is_student = True     # bool
print("Name:", name, "| Type:", type(name))
print("Age:", age, "| Type:", type(age))
print("Height:", height, "| Type:", type(height))
print("Student:", is_student, "| Type:", type(is_student))
```

```
age_float = float(age)
height_int = int(height)
print("\nAfter Conversion:")
print("Age as Float:", age_float, "| Type:", type(age_float))
print("Height as Integer:", height_int, "| Type:", type(height_int))
```

Q3.(Operators - Arithmetic)

```
a = 15
b = 4
print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Floor Division:", a // b)
print("Modulo:", a % b)
print("Exponentiation:", a ** b)
```

Q4.(Operators - Comparison & Logical)

```
x = 25
y = 30
z = 25
print("1. Is x greater than y?", x > y)
print("2. Is x equal to z?", x == z)
print("3. Is x <= y AND y != z?", (x <= y) and (y != z))
```

```
print("4. Is x > y OR x == z?", (x > y) or (x == z))
```

Q5.(Input/Output)# Input/Output Program

```
name = input("Enter your name: ")
birth_year = input("Enter your birth year: ")
birth_year = int(birth_year)
current_year = 2026
age = current_year - birth_year
print("\n---- User Information ----")
print("Name:", name)
print("Age:", age)
```

Q6.(Combined - Variables, Operators & Input)

```
height = float(input("Enter your height (in meters): "))
weight = float(input("Enter your weight (in kg): "))
bmi = weight / (height ** 2)
print("\n---- BMI Result ----")
print("Your BMI is:", round(bmi, 2))
```

Q7.(Comments)

This program calculates the area and perimeter of a rectangle.

The user provides the dimensions, and the program performs the calculations and displays the results.

```
length = float(input("Enter the length of the rectangle: "))
```

```
width = float(input("Enter the width of the rectangle: "))
```

```
area = length * width
```

```
perimeter = 2 * (length + width)
```

```
print("\n---- Rectangle Details ----")
```

```
print("Area =", area)
```

```
print("Perimeter =", perimeter)
```

```
print("\nCalculation completed successfully!")
```

Q8.(Operators - Assignment & Membership)

```
fruits = ["mango", "apple", "banana", "cherry"]
```

```
score = 50
```

```
score += 25
```

```
apple_exists = "apple" in fruits
```

```
grape_not_exists = "grape" not in fruits
```

```
print("Updated Score:", score)
```

```
print("Is 'apple' in the fruits list?", apple_exists)
```

```
print("Is 'grape' not in the fruits list?", grape_not_exists)
```

Q9.(Mini Project - All Concepts)

```
student_name = input("Enter your name: ")
student_age = int(input("Enter your age: "))
student_city = input("Enter your city: ")
subject1_marks = float(input("Enter marks for Subject 1: "))
subject2_marks = float(input("Enter marks for Subject 2: "))
subject3_marks = float(input("Enter marks for Subject 3: "))
total_marks = subject1_marks + subject2_marks + subject3_marks
maximum_marks = 300
percentage = (total_marks / maximum_marks) * 100
print("\n" + "=" * 35)
print("    STUDENT PROFILE")
print("=" * 35)
print("Name    :", student_name)
print("Age     :", student_age)
print("City    :", student_city)
print("\nMarks:")
print("Subject 1 :", subject1_marks)
print("Subject 2 :", subject2_marks)
print("Subject 3 :", subject3_marks)
print("\nTotal Marks:", total_marks)
print("Percentage :", round(percentage, 2), "%")
print("\n" + "=" * 35)
```

Q10.(Debugging & Understanding)

```
Name = "Alice"
```

```
age = 20
```

```
city = "Amsterdam"
```

```
print("My name is " + Name)
```

```
print("I am", age, "years old")
```

```
print("I live in", city)
```

```
print("Adult:", age > 18)
```